Design, Verification, and Synthesis of a Traffic Light Controller with Emergency Priority System Using Verilog HDL

Technical Project Report

RTL Design, Functional Simulation, Synthesis, and Resource Analysis using AMD Vivado

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Project Type

Academic RTL Design Project

Development Environment

Language	:	Verilog HDL
EDA Tool	:	AMD Vivado 2025.2
Verification	:	Vivado Simulator
Target Design	:	Traffic Light Controller with Emergency Priority
Architecture	:	Moore FSM Based Controller
Hardware Target	:	FPGA Synthesis Only (No Board Implementation)

July 2026

Abstract

Traffic management systems are essential for ensuring safe and efficient transportation in modern urban environments. Conventional traffic signal systems operate using fixed timing sequences and often lack mechanisms to provide immediate priority access to emergency vehicles such as ambulances, fire trucks, and police vehicles. This project presents the design, verification, and synthesis of a Traffic Light Controller with an Emergency Priority System using Verilog HDL.

The proposed controller is implemented as a Moore Finite State Machine (FSM) consisting of normal traffic operation states and dedicated emergency handling states. Under normal conditions, the controller follows a predefined sequence of traffic light transitions between two intersecting roads. Upon detection of an emergency vehicle request, the controller safely overrides the normal sequence by transitioning through an appropriate yellow signal state before granting immediate right-of-way to the emergency route, thereby ensuring both safety and reduced response time.

The design was developed and verified using AMD Vivado 2025.2. Functional verification was performed using a comprehensive testbench covering normal operation, emergency requests on both roads, simultaneous emergency conditions, and reset behavior. Synthesis results demonstrate that the controller requires minimal hardware resources, utilizing only 20 Look-Up Tables (LUTs) and 8 Flip-Flops (FFs), indicating a lightweight and efficient implementation suitable for hardware realization on FPGA platforms. Power analysis further confirms the low complexity and energy-efficient nature of the design.

The project demonstrates the application of RTL design methodologies, FSM-based control logic, simulation-driven verification, and synthesis analysis in digital system design using Verilog HDL.

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Chapter 1

Introduction and Methodology

1.1 Introduction

Traffic signal control systems are essential for ensuring safe and efficient vehicular movement at road intersections. Conventional traffic controllers generally operate using fixed timing sequences and often lack mechanisms to provide immediate priority access to emergency vehicles such as ambulances, fire engines, and police vehicles.

To address this limitation, this project presents the design of a Traffic Light Controller with an Emergency Priority System using Verilog HDL. The controller manages traffic flow at a two-road intersection under normal operating conditions while allowing emergency vehicles on either road to temporarily override the normal traffic sequence and obtain priority access.

The design is implemented using a Moore Finite State Machine (FSM) consisting of normal traffic states and dedicated emergency handling states. Functional verification, synthesis, resource utilization analysis, and power estimation were performed using AMD Vivado 2025.2.

Although hardware implementation on an FPGA board was not performed due to the unavailability of hardware resources, the design was thoroughly validated through simulation and synthesis reports, following an RTL design workflow similar to that used in FPGA and ASIC development.

1.2 Problem Statement

Conventional traffic light controllers operate using fixed timing sequences and do not provide mechanisms for prioritizing emergency vehicles at intersections. This can increase emergency response times and reduce traffic management efficiency. Therefore, a traffic control system capable of temporarily overriding normal operation to provide emergency vehicle priority while maintaining safe signal transitions is required.

1.3 Objectives

- Design a traffic light controller using Verilog HDL.
- Implement emergency vehicle priority using a Moore FSM.
- Verify functionality through simulation.

- Analyze synthesis, resource utilization, and power consumption.

1.4 Scope of the Project

The scope of the project is limited to the RTL design and verification of a two-road traffic intersection controller with emergency priority support. The project includes simulation, synthesis, utilization analysis, and power estimation but excludes physical FPGA implementation and real-world sensor interfacing. Future enhancements may include adaptive traffic timing, traffic density estimation, pedestrian crossing support, and integration with hardware sensors for real-time deployment.

1.5 Design Methodology

The development of the Traffic Light Controller with Emergency Priority System followed a structured RTL design methodology commonly adopted in digital system design and FPGA development workflows. The implementation process began with requirement analysis and proceeded through FSM design, RTL coding, simulation-based verification, synthesis, and implementation analysis.

The overall design methodology adopted in this project is illustrated in Figure 1.1.

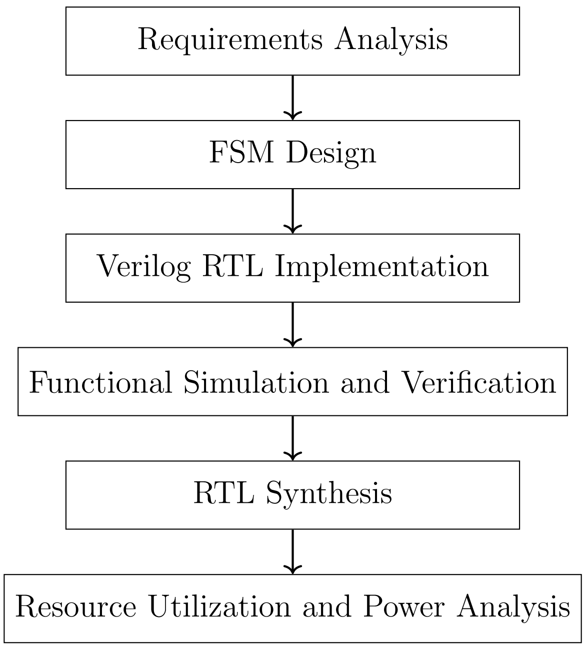


Figure 1.1: Project Development Methodology

1.5.1 Requirements Analysis

The first stage involved identifying the functional requirements of the controller. The system was required to:

- Control traffic flow at a two-road intersection.
- Follow predefined timing intervals for each traffic signal state.

- Support emergency vehicle priority requests from both roads.
- Ensure conflict-free operation by preventing simultaneous green signals.
- Transition safely through yellow states before changing traffic directions.

1.5.2 FSM Design

The traffic controller was modeled using a Moore Finite State Machine (FSM). Six states were defined to represent normal traffic operation and emergency conditions. The FSM architecture provides deterministic operation and simplifies the implementation of priority override mechanisms.

1.5.3 RTL Implementation

The controller was implemented using Verilog HDL following a modular RTL design methodology. The implementation consists of a state register, counter logic, next-state combinational logic, and output decoding logic.

1.5.4 Simulation and Verification

Functional verification was performed using the Vivado Simulator. A comprehensive testbench was developed to verify reset functionality, normal traffic operation, emergency request handling for both roads, simultaneous emergency requests, and recovery to normal operation after emergency clearance.

1.5.5 Synthesis and Analysis

Following successful verification, the RTL design was synthesized using AMD Vivado 2025.2. The synthesis process generated RTL schematics, synthesized netlists, resource utilization reports, and power estimation reports which were used to evaluate the hardware efficiency of the design.

Chapter 2

System Design and Architecture

2.1 System Overview

The proposed Traffic Light Controller is designed to regulate traffic flow at a two-road intersection while providing priority access to emergency vehicles. Under normal operating conditions, the controller follows a predefined sequence of traffic light transitions between the two roads. Upon detection of an emergency request, the controller temporarily overrides the normal traffic sequence and grants immediate access to the corresponding road while maintaining safe signal transitions.

The controller is implemented using a Moore Finite State Machine (FSM) and consists of normal traffic operation states and dedicated emergency handling states.

2.2 Input and Output Specifications

2.2.1 Input Signals

Table 2.1: Input Signal Description

Signal	**Width**	**Description**
clk	1	System clock signal
reset	1	Active high asynchronous reset signal
emg_A	1	Emergency request input for Road A
emg_B	1	Emergency request input for Road B

2.2.2 Output Signals

Table 2.2: Output Signal Description

Signal	**Width**	**Description**
A_red	1	Red light output for Road A
A_yellow	1	Yellow light output for Road A
A_green	1	Green light output for Road A
B_red	1	Red light output for Road B
B_yellow	1	Yellow light output for Road B
B_green	1	Green light output for Road B

2.3 System Architecture

The controller consists of four major functional blocks:

- Counter and timing logic.
- FSM controller.
- Emergency request handling logic.
- Output decoding logic.

The overall architecture of the proposed controller is shown in Figure 2.1.

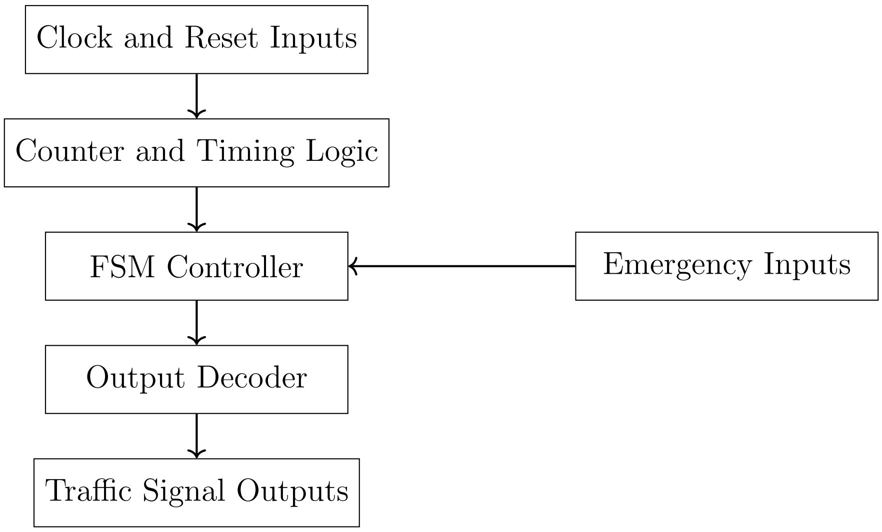


Figure 2.1: System Architecture of the Traffic Light Controller

2.4 Finite State Machine Design

The controller is implemented using a Moore FSM consisting of six states:

Table 2.3: FSM State Description

State	**Encoding**	**Description**
S0	000	Road A Green, Road B Red
S1	001	Road A Yellow, Road B Red
S2	010	Road A Red, Road B Green
S3	011	Road A Red, Road B Yellow
S4	100	Emergency State for Road A
S5	101	Emergency State for Road B

Figure 2.2 illustrates the state transitions of the controller.

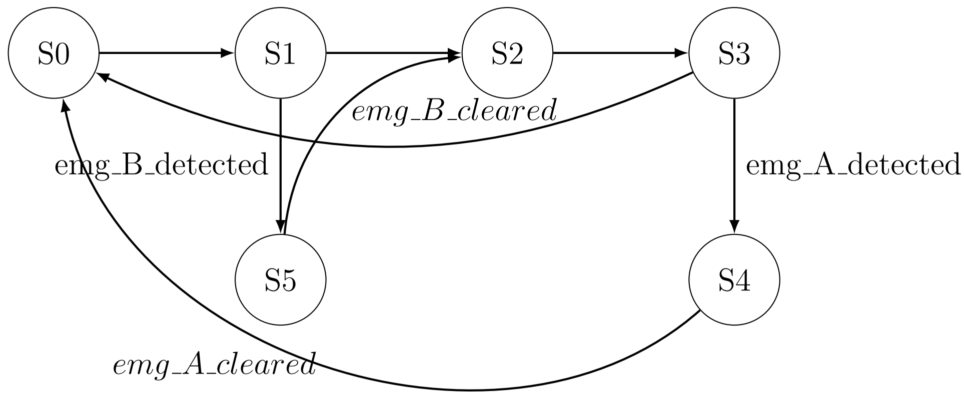


Figure 2.2: Finite State Machine Representation of the Traffic Light Controller with Emergency Priority Support

2.5 Emergency Priority Mechanism

The emergency priority mechanism allows the controller to temporarily override normal operation and grant immediate access to the road requesting emergency priority.

The controller follows the sequence:

- Detect emergency request.
- Complete safe transition through yellow state.
- Grant green signal to the requested road.
- Maintain priority until emergency request is cleared.
- Resume normal operation.

The emergency handling process is illustrated in Figure 2.3.

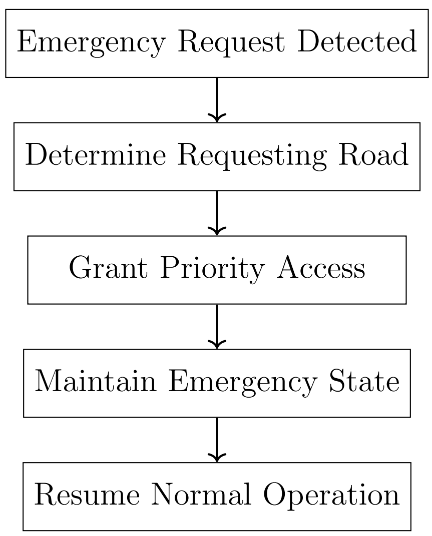


Figure 2.3: Emergency Priority Handling Flow

2.6 Timing Parameters

The timing values used during simulation are shown in Table 2.4.

Table 2.4: Traffic Signal Timing Parameters

State	**Duration (Clock Cycles)**
Road A Green	10
Road A Yellow	3
Road B Green	10
Road B Yellow	3

Chapter 3

Verification and Synthesis Analysis

3.1 Verification Methodology

Functional verification was performed using the AMD Vivado Simulator to ensure correct operation of the traffic controller under both normal and emergency conditions. A comprehensive testbench was developed to validate state transitions, emergency priority handling, and reset functionality.

The verification process adopted in this project is illustrated in Figure 3.1.

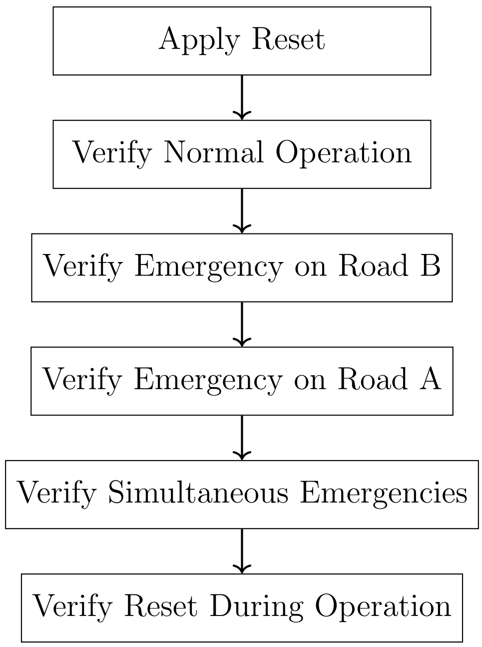


Figure 3.1: Verification Methodology

3.2 Testbench Description

The verification environment was developed using a self-contained Verilog testbench. The testbench generates the system clock, applies reset conditions, and stimulates the emergency request inputs under different operating scenarios.

The following verification cases were considered:

Table 3.1: Verification Test Cases

Test Case	Description	Result
TC1	Reset functionality	Pass
TC2	Normal traffic operation	Pass
TC3	Emergency request on Road B	Pass
TC4	Emergency request on Road A	Pass
TC5	Simultaneous emergency requests	Pass
TC6	Reset during operation	Pass

3.3 Simulation Results

The behavioral simulation waveform obtained from the verification process is shown in Figure 3.2.

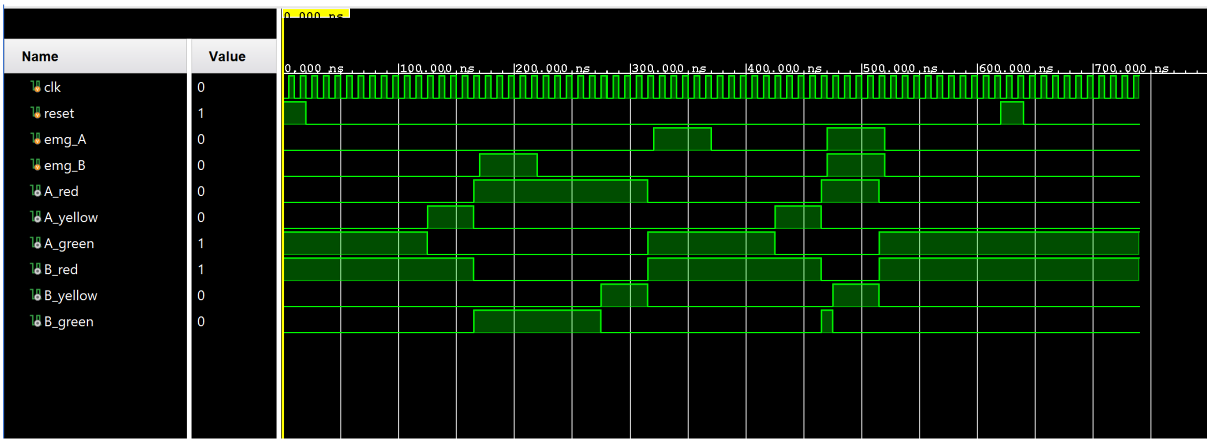


Figure 3.2: Behavioral Simulation Waveform

The simulation results confirm correct state transitions, proper emergency handling behavior, and conflict-free traffic signal operation.

3.4 RTL Schematic

Following successful verification, the design was synthesized using AMD Vivado 2025.2. Figure 3.3 shows the RTL schematic generated from the Verilog source code.

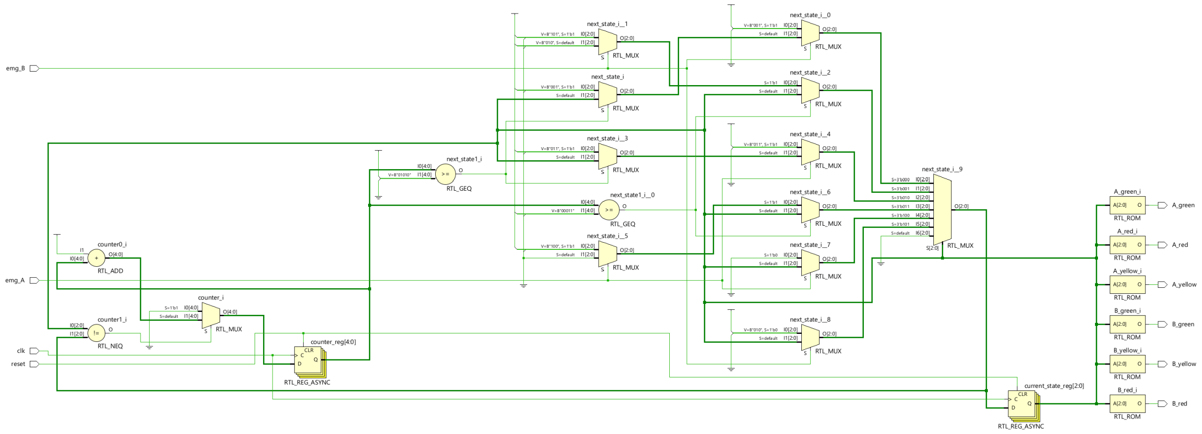


Figure 3.3: RTL Schematic of the Traffic Light Controller

The RTL schematic confirms the expected architecture consisting of state registers, counter logic, combinational next-state logic, and output decoding circuitry.

3.5 Synthesized Netlist

The synthesized implementation generated by Vivado is shown in Figure 3.4.

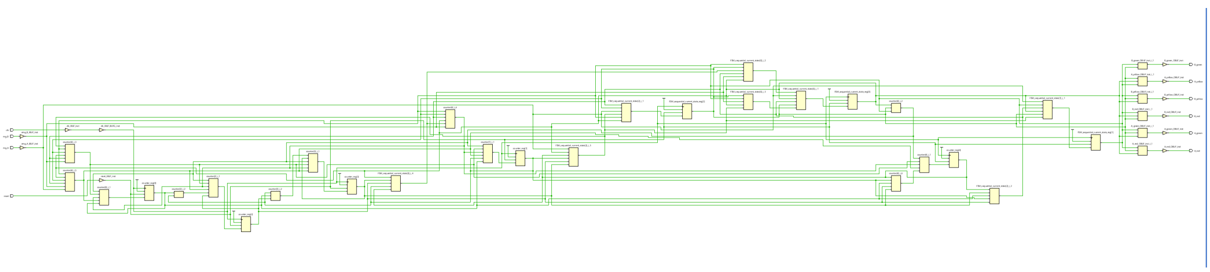
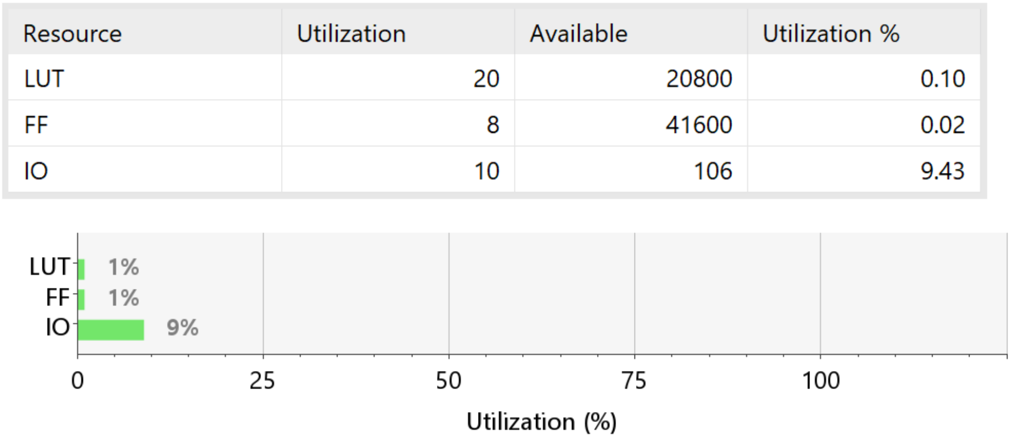


Figure 3.4: Synthesized Netlist of the Traffic Light Controller

The synthesized design confirms successful mapping of the RTL description into target FPGA resources during synthesis.

3.6 Resource Utilization Analysis

The hardware resource utilization report obtained after synthesis is shown in Figure 3.5.



Resource	Utilization	Available	Utilization %
LUT	20	20800	0.10
FF	8	41600	0.02
IO	10	106	9.43

Figure 3.5: Resource Utilization Report

The synthesized design utilizes only 20 Look-Up Tables (LUTs) and 8 Flip-Flops (FFs), demonstrating an efficient and lightweight implementation suitable for hardware realization on FPGA platforms.

Table 3.2: Post-Synthesis Resource Utilization

Resource	Used	Available	Utilization
LUT	20	20800	0.10%
FF	8	41600	0.02%
I/O	10	106	9.43%

3.7 Power Analysis

The estimated power consumption of the synthesized design is shown in Figure 3.6.

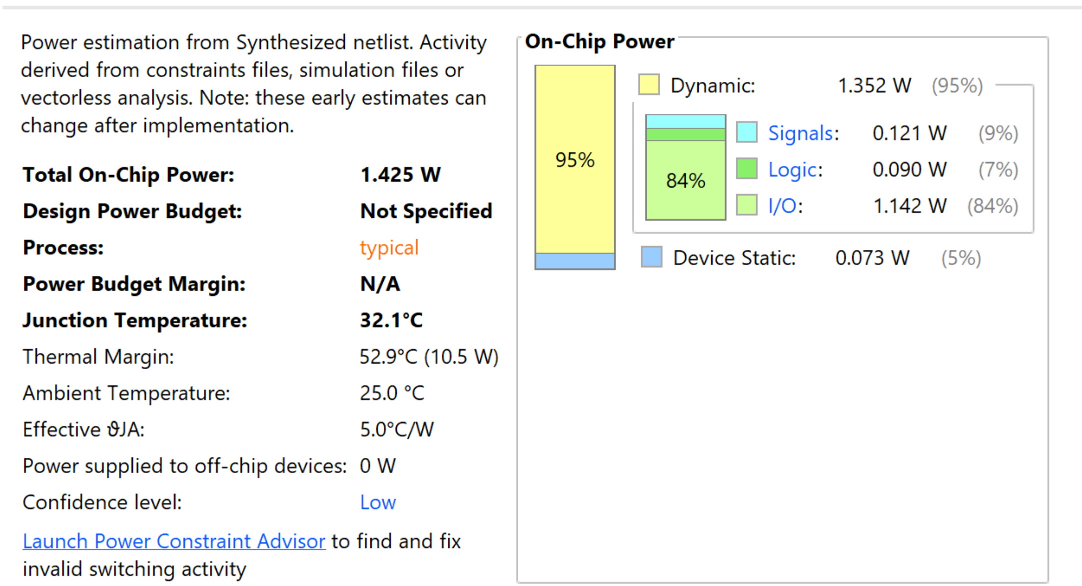


Figure 3.6: Power Analysis Report

The total estimated on-chip power consumption is 1.425 W, with the majority of dynamic power attributed to I/O activity. The low logic power consumption reflects the simplicity and efficiency of the FSM-based architecture.

Table 3.3: Power Summary

Parameter	**Value**
Total On-Chip Power	1.425 W
Dynamic Power	1.352 W
Static Power	0.073 W
Junction Temperature	32.1°C

Conclusion

In this project, a Traffic Light Controller with an Emergency Priority System was successfully designed, verified, and synthesized using Verilog HDL and AMD Vivado 2025.2. The controller was implemented using a Moore Finite State Machine (FSM) architecture to regulate traffic flow at a two-road intersection while providing priority access to emergency vehicles.

The design was functionally verified through simulation under various operating conditions including normal traffic operation, emergency requests on both roads, simultaneous emergency scenarios, and reset conditions. Simulation results confirmed correct state transitions and conflict-free signal operation.

Synthesis results demonstrated that the proposed controller requires minimal hardware resources, utilizing only 20 LUTs and 8 Flip-Flops, indicating an efficient and lightweight implementation. Power analysis further showed low logic power consumption due to the simplicity of the FSM-based architecture.

Although hardware implementation on an FPGA board was not performed due to the unavailability of hardware resources, the project successfully demonstrated a complete RTL design flow including design entry, simulation, synthesis, resource utilization analysis, and power estimation. The methodologies used in this work closely resemble industrial FPGA and ASIC design practices.

Future improvements may include adaptive traffic control based on traffic density, pedestrian crossing support, real-time sensor interfacing, and deployment on an FPGA platform for hardware validation.

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